📖 1. Room Overview

The **Offensive Security Info** room provides a high-level introduction to the "Red Team" side of cybersecurity. While defensive security focuses on protection, **Offensive Security** focuses on identifying vulnerabilities by thinking like an attacker to strengthen a system's overall posture.

⚔️ 2. What is Offensive Security?

Offensive security is a proactive and adversarial approach to protecting computer systems, networks, and individuals from attacks. It involves several specialized career paths:

🕵️ Penetration Tester (Ethical Hacker)

The most common role in the industry.

- **Goal:** To legally "hack" into a network or application to find vulnerabilities before a real criminal does.
- **Method:** Following a structured process: **Information Gathering -> Scrutiny/Scanning -> Exploitation -> Post-Exploitation.**

🔴 Red Teaming

A more advanced, long-term simulation.

- **Goal:** To test not just the software, but the **people and processes** of an organization.
- **Activity:** Using "social engineering" (phishing) or physical intrusion to see if the SOC (Security Operations Center) can detect them.

🏗️ 3. The Offensive Mindset: Digital vs. Physical

To be successful in offensive security, one must understand that vulnerabilities aren't always technical; they can be physical or human-centric.

- **Software Vulnerabilities:** Finding "bugs" in code, such as **SQL Injection** or **Buffer Overflows**.
- **Physical Security:** Testing locks, cameras, and badge readers to gain entry to a server room.
- **Social Engineering:** Manipulating individuals into giving up passwords or clicking malicious links.

🛠️ 4. Essential Tools of the Trade

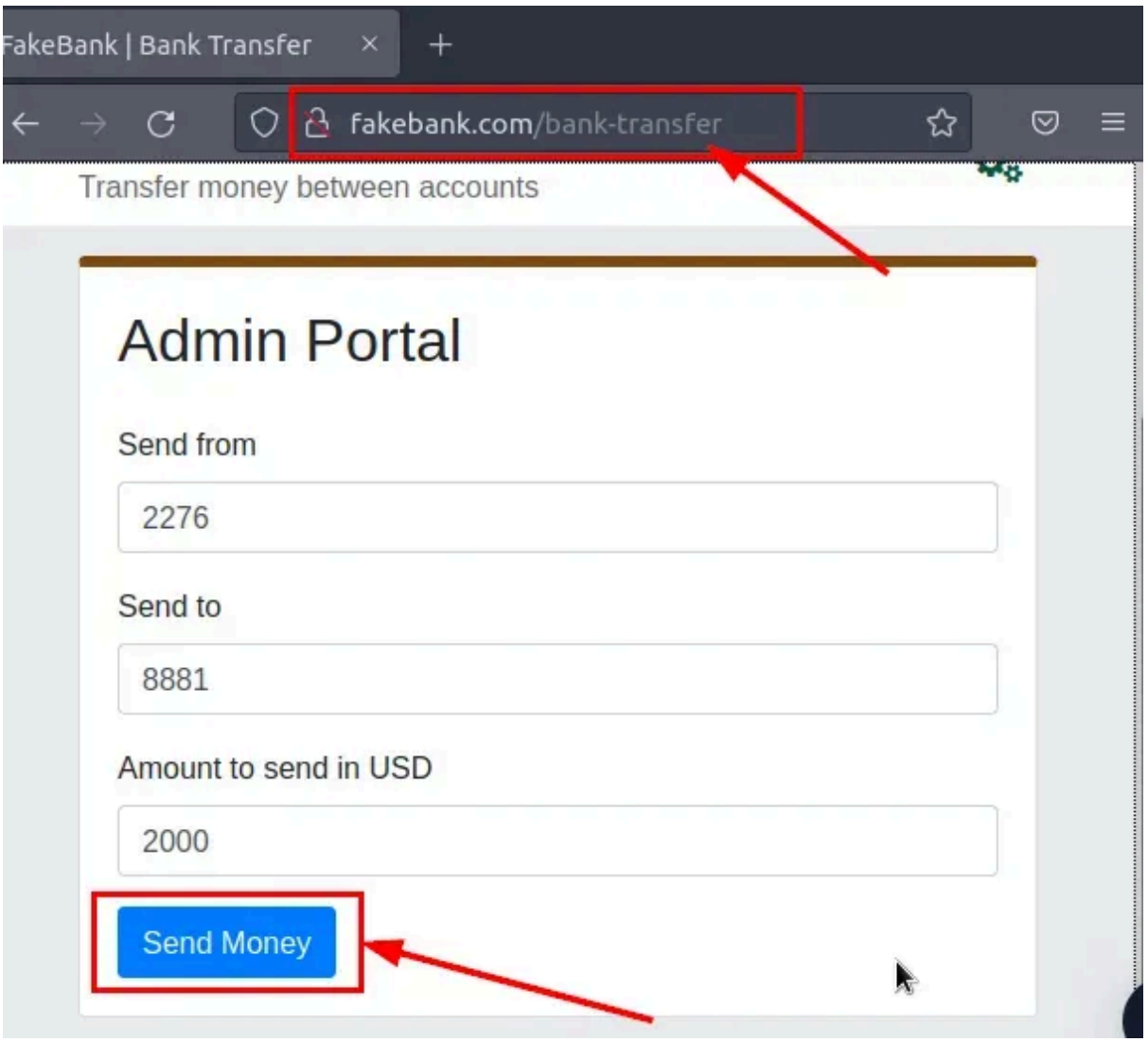
During this lab, I was introduced to the primary tools used by offensive security professionals:

- **Nmap:** Used for network discovery and security auditing.
- **Burp Suite:** The industry standard for web application security testing.
- **Metasploit:** A powerful framework used to find, exploit, and validate vulnerabilities.
- **John the Ripper:** A fast password cracker used to test the strength of user credentials.

🚩 5. Lab Activity: My First "Hack"

In this task, I used a simulated terminal to perform a basic exploitation and retrieve a hidden flag. This demonstrated how a small oversight in a web application can lead to total system compromise.

📸 Lab Evidence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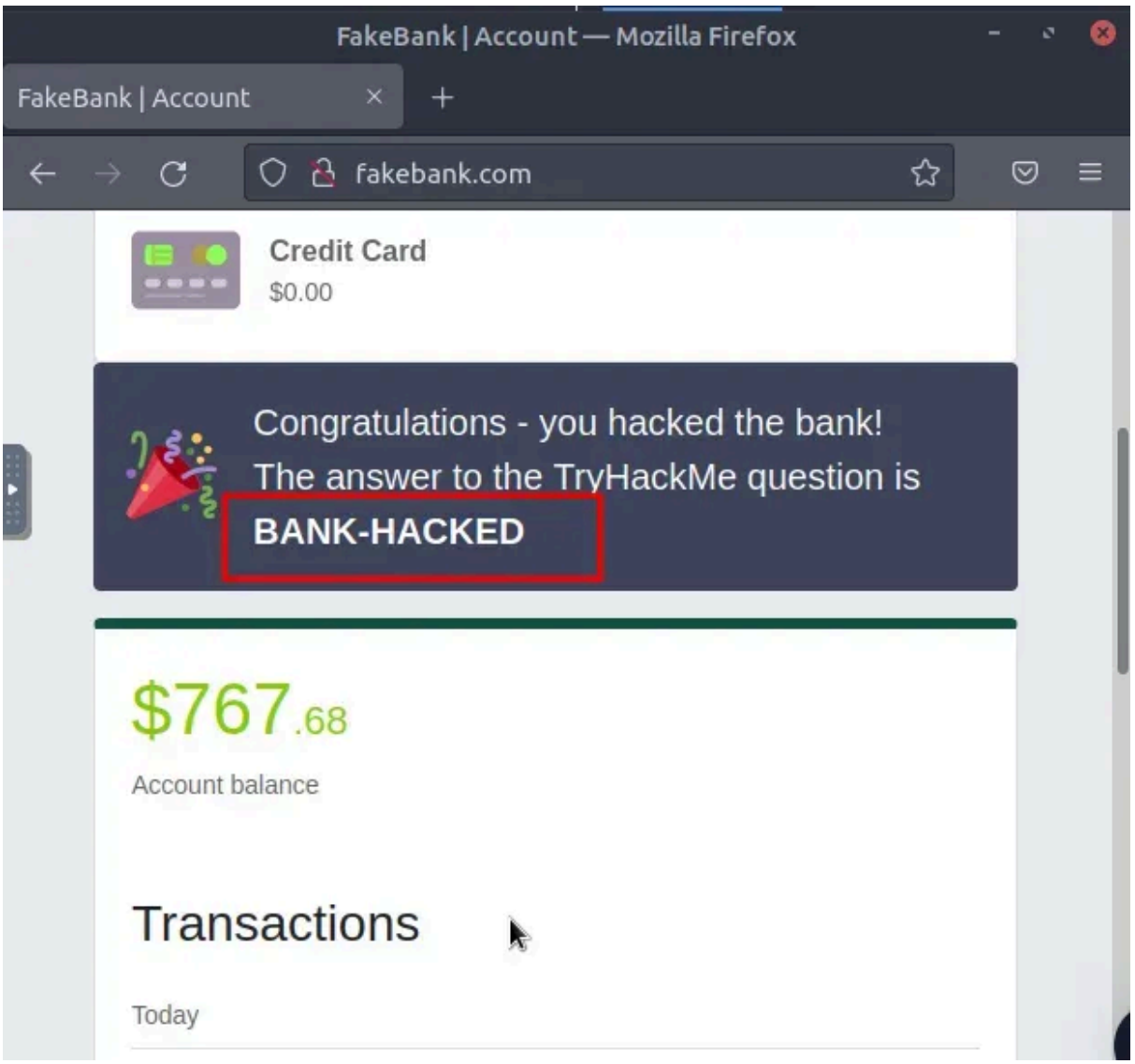
fakebank.com/bank-transfer
Transfer money between accounts
Admin Portal
Send from
2276
Send to
8881
Amount to send in USD
2000
Send Money



🎓 6. Conclusion & Skills Gained

By completing this module, I gained a professional foundation in:

- Understanding the **Ethical Hacking** legal and moral framework.
- Identifying the core differences between **Vulnerability Scanning** and **Penetration Testing**.
- Recognizing the standard **Red Team** lifecycle.